Bangladesh University of Engineering and Technology

Department of Electrical and Electronic Engineering



Project Report

Design of State Variable Filter Using PSpice

Course No: EEE 208 (S)

Course Title: Electronics Circuits II Laboratory

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Objective:

(1) To design and simulate state variable filter circuits using PSpice and to produce high pass, band pass and low pass response from different terminals of the same circuit.

(2) To plot output characteristics of filter circuits such as output voltage, gain, dB gain in frequency domain analysis setup.

(3) To show that the three main filter parameters such as gain, Q-factor and corner frequency can be controlled independently by simply changing one or two resistor values while keeping the design of the rest of the circuit fixed.

Introduction:

A state variable circuit is a type of active filter design which uses multiple feedbacks and integrators to produce all types of filter response such as high pass, band pass, low pass and band reject or notch from a single circuit. It is very useful to provide a specific Q factor, corner frequency and gain.

Based on the number of op-amps used, state variable filter can be of three types-

(1) 2 Op-amp configuration (produces high pass and low pass only)

(2) 3 Op-amp configuration (produces high pass, band pass and low pass)

(3) 4 Op-amp configuration (can produce notch, high pass, band pass and low pass)

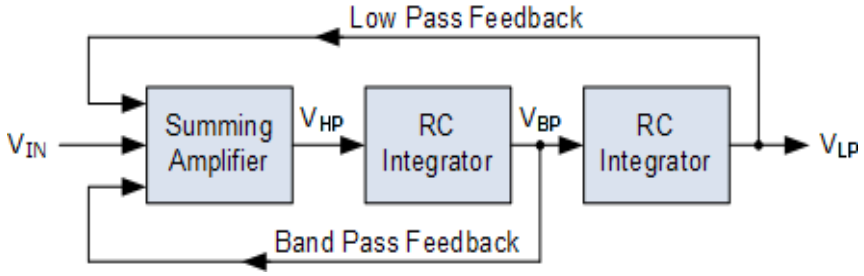


Fig 1: Block Diagram of a Standard 3 Op-Amp State Variable Filter

A standard 3 op-amp state variable filter is a second order filter. It has 2 integrators (hence the filter is second order; each integrator is first order circuit because they consist of one energy element/capacitor) and a summing amplifier. Output of the summing amplifier is the high pass signal which goes through the integrators and become band pass and low pass signals consecutively. Finally, low pass signal and

a portion of band pass signal are fed back to summing amplifier where they are joined by the input signal. Together, they make the high pass signal at the output.

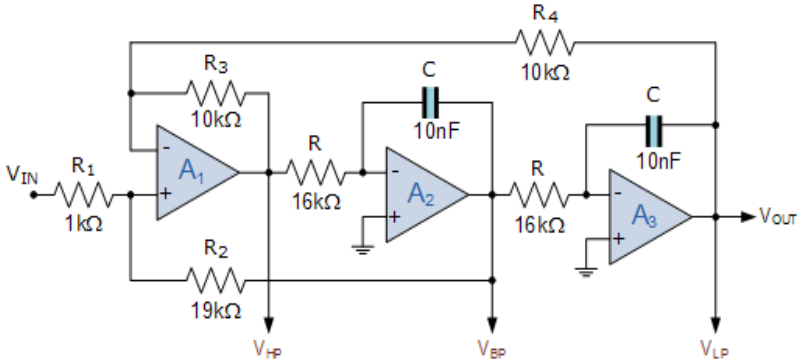


Fig 2: Schematic Diagram of a sample State Variable Filter

Second order state variable filter's high pass and low pass signals have 40 dB/decade and -40 dB/decade slope respectively at the roll-off while the band pass signal have 20 dB/decade and -20 dB/decade slope respectively at before and after the center frequency.

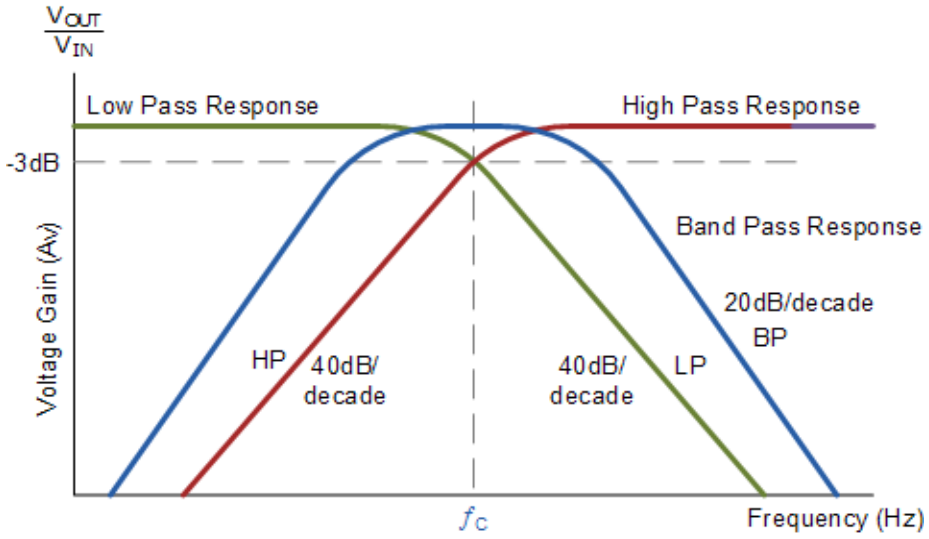


Fig 3: Voltage Gain (dB) vs Frequency Graph of a 2nd Order State Variable Filter

In the 4 op-amp configuration of second order state variable filter, we can get four types of responses simultaneously. An extra op-amp added in the beginning produces the notch response. Alternatively, we can get the notch signal by simply adding high pass and low pass signal through an adder.

The 2 op-amp version of state variable filter has a summing amplifier and only one integrator. That's why it is called a first order filter and has a 20 dB/decade slope at

roll-off for high pass response as well as -20 dB/decade slope for low pass response. Bandpass or band reject response can't be achieved in this version.

State variable filter (also known as Universal Filter) has some great advantages over other basic filter designs. Firstly, all types of filter responses can be realized from a single circuit without any complication. Secondly, filter parameters like gain, Q factor and corner frequency can be adjusted independently.

Background Theory:

❖ How different filter responses are produced in the circuit:

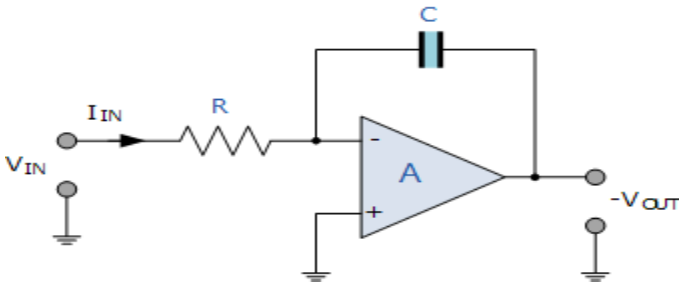


Fig 4: An Integrator Circuit

Input/output relation in time domain:

$$V_{OUT} = \frac{-1}{RC}\int_0^t V_{IN}\, dt$$

Input/output relation in frequency domain:

$$V_{OUT} = \frac{-1}{2\pi f RC}\, V_{IN}$$

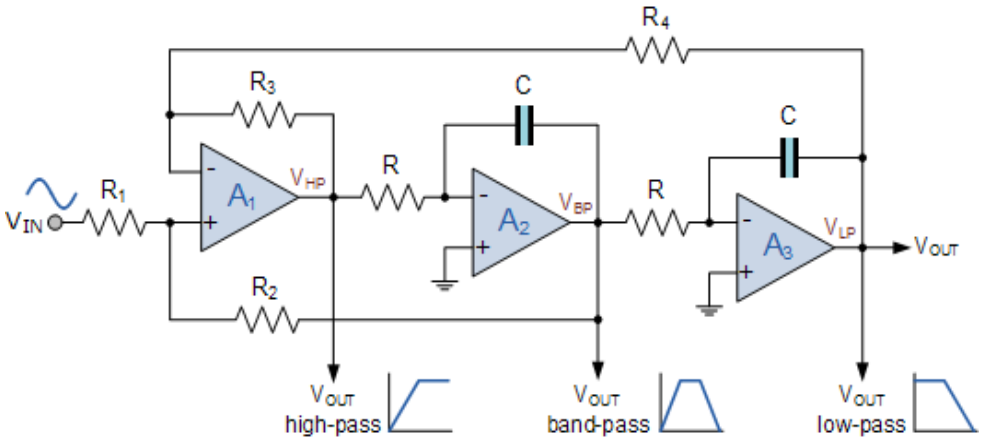


Fig 5: A 3 Op-amp State Variable Filter (Input going into non-inverting terminal)

Here, A2 and A3 are integrators and A1 is summing amplifier.

Input/output relation for A2:

$V_{BP} = \frac{-1}{2\pi fRC} V_{HP}$

Input/output relation for A3:

$V_{LP} = \frac{-1}{2\pi fRC} V_{BP}$

$\Rightarrow V_{LP} = \frac{-1}{2\pi fRC} * (\frac{-1}{2\pi fRC} V_{HP}) = \frac{1}{(2\pi fRC)^2} V_{HP}$

So, band pass signal is produced by integrating high pass signal once and low pass signal is produced by integrating high pass signal twice.

Input/Output Relation for A1:

[Reference: Fig- 05, Page- 04]

$\frac{V_{IN}-(V+)}{R_1} = \frac{(V+)-V_{BP}}{R_2}$

$\Rightarrow (V+) = \frac{V_{IN}*R_2+V_{BP}*R_1}{R_1+R_2}$ .....(1)

Again, $\frac{V_{HP}-(V-)}{R_3} = \frac{(V-)-V_{LP}}{R_4}$

$\Rightarrow (V-) = \frac{V_{HP}*R_4+V_{LP}*R_3}{R_3+R_4}$.....(2)

From (1) and (2),

(V+) = (V-)

$\Rightarrow V_{HP} = V_{IN} \times \frac{R_2(R_3+R_4)}{R_4(R_1+R_2)} + V_{BP} \times \frac{R_1(R_3+R_4)}{R_4(R_1+R_2)} - V_{LP} * \frac{R_3}{R_4}$

$R_3 = R_4$ is considered for easier control over circuit. In that case,

$$\mathbf{V_{HP} = V_{IN} * \frac{2*R_2}{R_1+R_2} + V_{BP} * \frac{2*R_1}{R_1+R_2} - V_{LP}}$$

So, the high pass signal is produced by adding input signal to band pass and then subtracting low pass signal.

Till now, we have explained through mathematical equations. These can also be explained by graphical analysis.

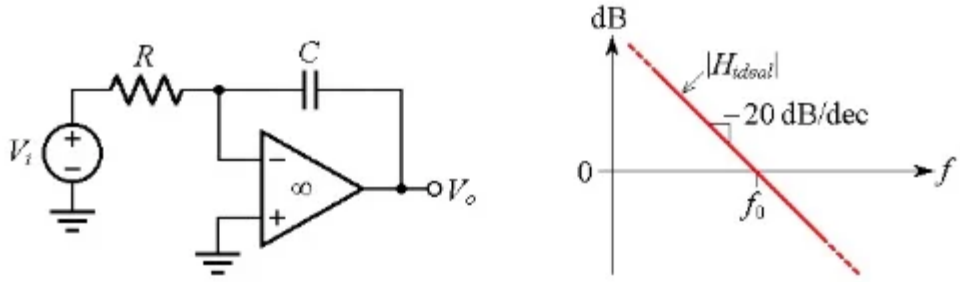


Fig 6: Ideal Response of an Integrator Circuit in Frequency Domain

So, if a signal passes through an integrator, **its slope will be reduced by 20 dB/decade.**

If we consider that high pass response is applied at the input of A2, we can make a logical assumption that its slope will be reduced by 20 dB/decade.

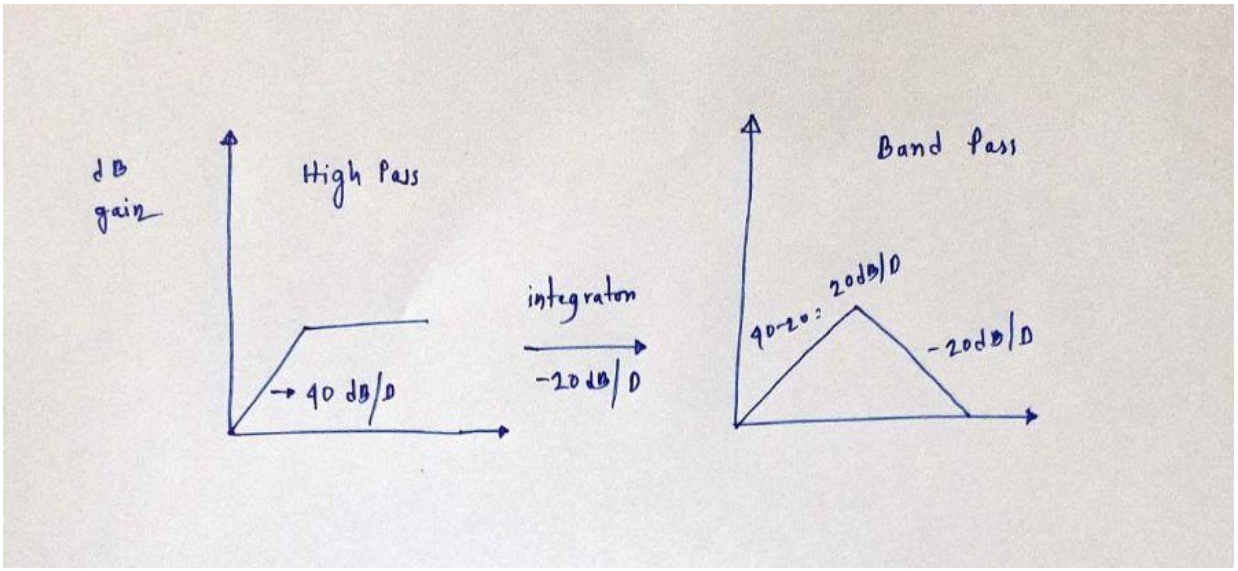


Fig 7: Band Pass Signal Production

First part of high pass response has 40 dB/decade slope which reduces down to 20 dB/decade and the second part of high pass signal has 0 dB/decade slope which becomes -20 dB/decade after passing through integrator. As can be seen from the figure, the output is a band pass response.

Similarly, if this band pass goes through A3, its slope will be reduced again by 20 dB/decade.

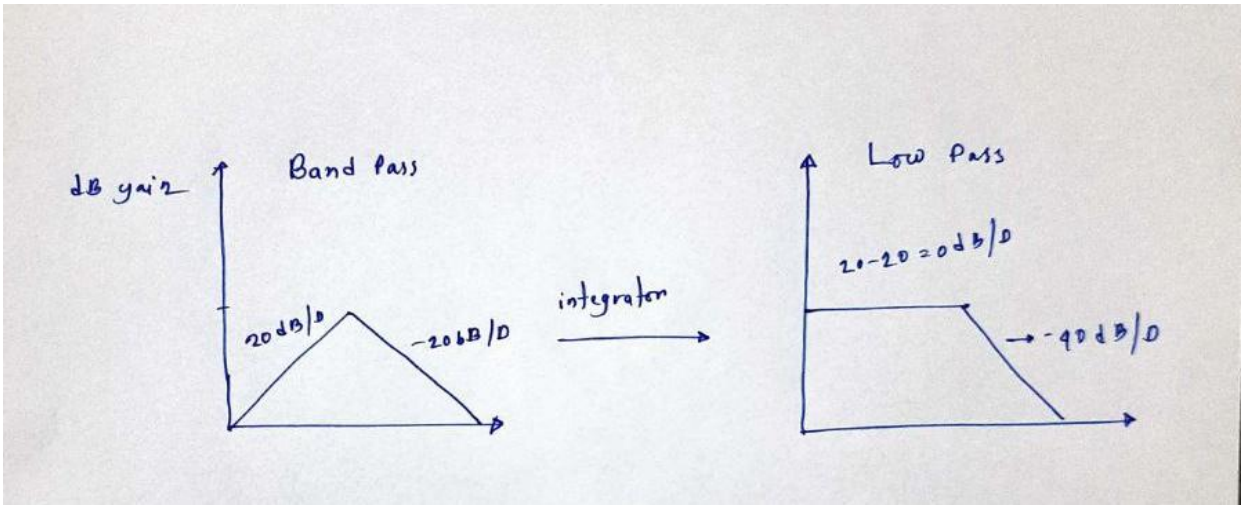


Fig 8: Low Pass Signal Production

First part of band pass response has 20 dB/decade slope which reduces to 0 dB/decade and the second part of high pass signal has -20 dB/decade slope which becomes -40 dB/decade after passing through integrator A3. As can be seen from the figure, the output is a low pass response.

Later, low pass and band pass feedback will be routed back to first op-amp A1 along with input. Output of this summing amplifier will be the high pass signal which we have considered in the first place and the loop will continue.

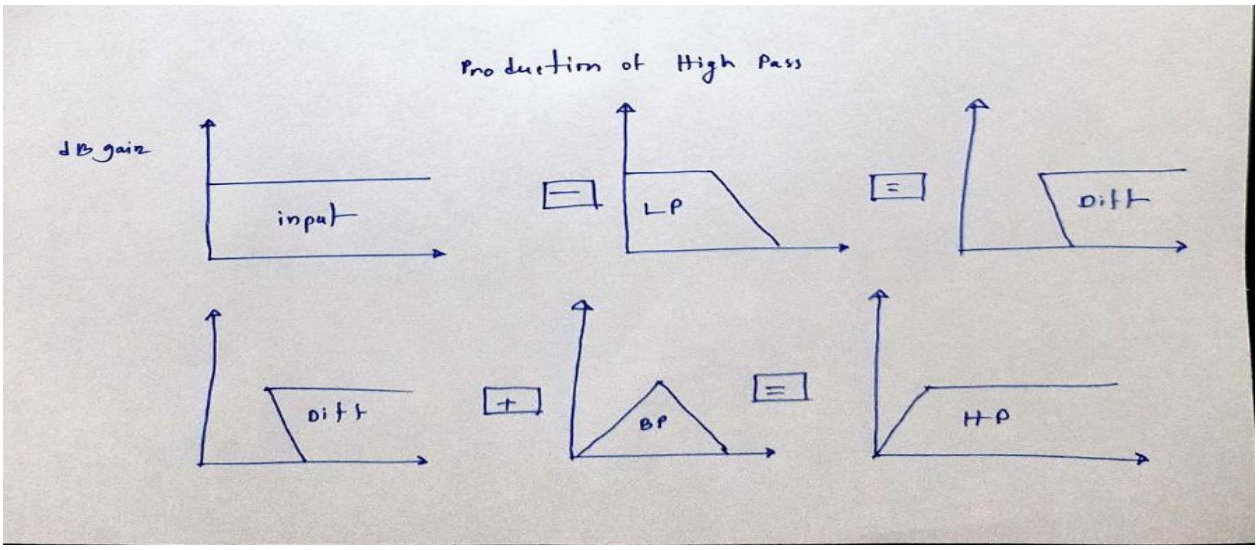


Fig 9: High Pass Signal Production

Here, low pass signal is routed back to inverting terminal and band pass signal is fed back into non-inverting terminal where it is joined by input signal. So, low pass signal will be subtracted from the input and the difference will be added to a small

portion of the band pass signal. As can be seen from the figure, the output is a high pass response.

2 op-amp and 4 op-amp versions of state variable filters has almost similar characteristics.

❖ Determining the Formula for Corner Frequency, Gain and Q factor:

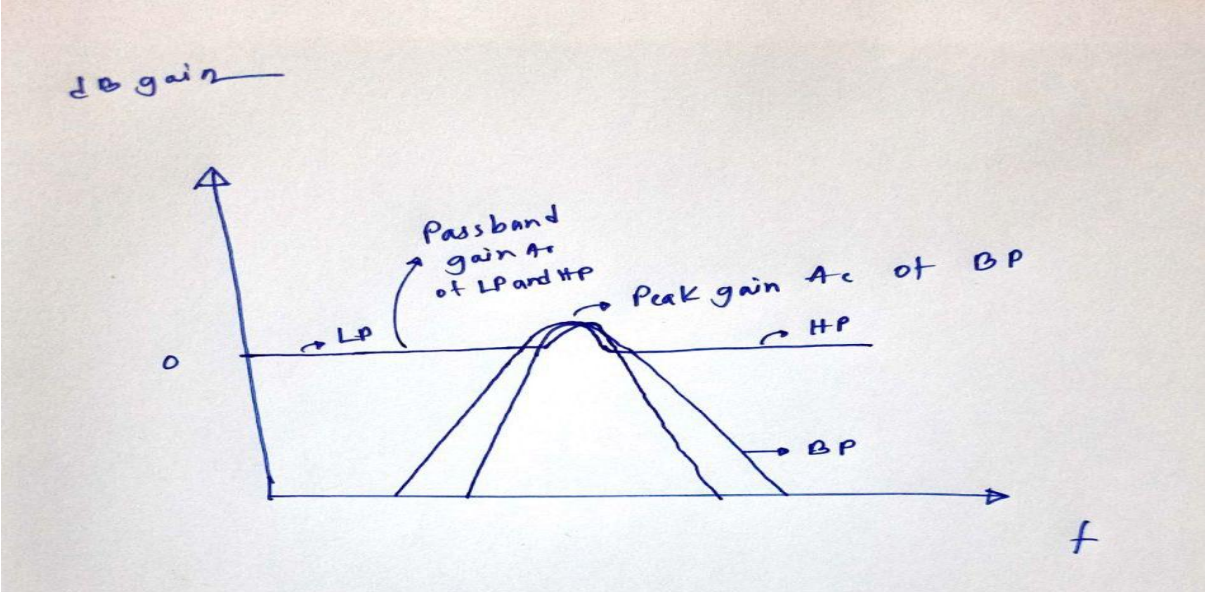


Fig 10: State Variable Filter Bode Plot

There are two types of gain present in the graph. One is the passband gain A_O of the high pass and the low pass signal. Another is the peak gain A_C (found at center frequency) of the band pass signal. They are related with the Q factor.

$A_C = A_O * Q$

The Q factor mainly depends on the bandwidth of the band pass response. Increasing the Q factor of the band pass signal will increase the peak gain A_C because gain-bandwidth product must remain constant. But a well-designed state variable filter will keep the passband gain (A_O) of the high pass and low pass response unchanged in the face of Q variation. That basically means changing Q factor of band pass signal will move the peak gain A_C up and down along y axis while passband gain of high pass and low pass will stay fixed at its position.

Now, State Variable Filter Transfer Function is -

$$\frac{V_{OUT}}{V_{IN}} = \frac{V_{LP}}{V_{IN}} = \frac{\frac{R_2(R_3+R_4)}{R_3(R_1+R_2)} \times \left(\frac{1}{RC}\right)}{\frac{R_3}{R_{4RC}} + \left(\frac{R_1(R_3+R_4)}{R_4(R_1+R_2)} \times \frac{1}{2\pi fRC}\right) + \left(\frac{1}{2\pi fRC}\right)^2}$$ **[Reference: Fig- 05, Page- 04]**

After normalizing-

$$\frac{V_{OUT}}{V_{IN}} = \frac{A_O\left(\frac{f}{f_0}\right)}{\left[1+2\varsigma\frac{f}{f_0}+\left(\frac{f}{f_0}\right)^2\right]}$$

Formula for corner frequency:

Same resistor and capacitor are used in both integrators to reduce the complications of calculation. In that case, the formula of corner frequency gets so much easier which is –

$$f_C = \frac{1}{2\pi RC} \text{ Hz}$$

All filters will have the same corner frequency f_C which doesn't depend on Q or gain. This frequency will remain constant as long as R and C values are unchanged.

Formula for Q-factor:

$$Q = \frac{f_C}{BW} = \frac{R_1(R_3+R_4)}{R_4(R_1+R_2)}\sqrt{\frac{R_3}{R_4}}$$ **[Reference: Fig- 05, Page- 04]**

If $R_3 = R_4$, then the formula reduces to $Q = \frac{2*R_1}{R_1+R_2}$

Formula for passband gain of high pass and low pass filter:

$$A_O = \frac{R_2(R_3+R_4)}{R_3(R_1+R_2)}$$ **[Reference: Fig- 05, Page- 04]**

If $R_3 = R_4$, then the formula reduces to $A_O = \frac{2*R_2}{R_1+R_2}$

- **Some useful observation:**

(1) Most of the resistances except two have been set to equal value for easier formula and filter control. Those two resistors will be varied to change the gain and Q-factor.

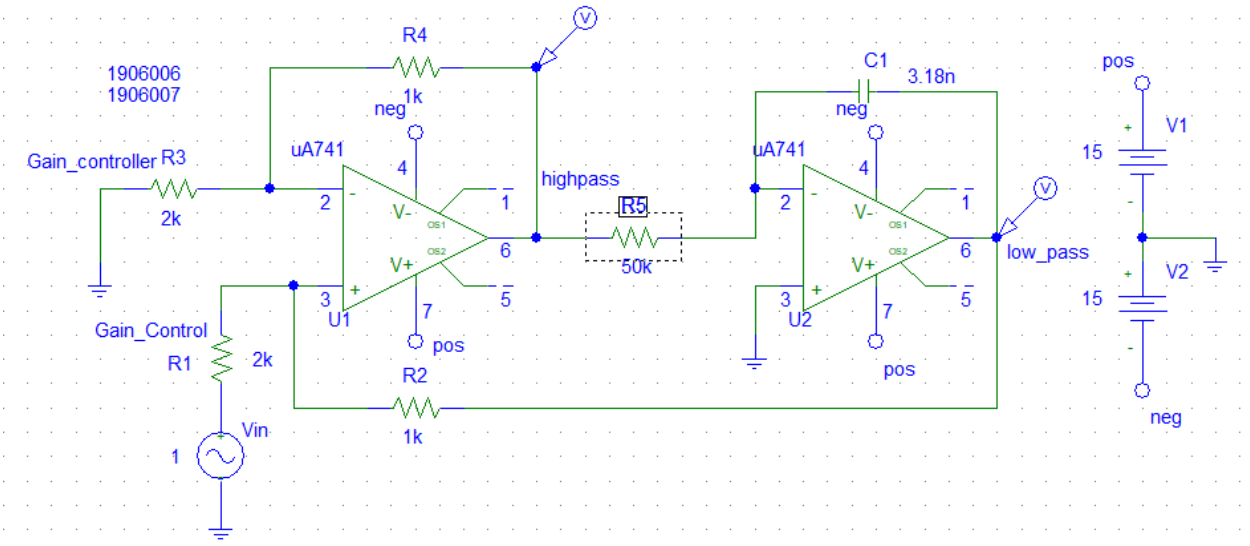
(2) Corner frequency is independent of gain or Q factor. So, it can be changed keeping other two fixed and vice versa.

(3) Changing passband gain can change Q factor while changing Q factor slightly changes the passband gain in 3 op-amp configuration. But passband gain and Q factor is completely independent in 4 op-amp configuration. So, using 4 op-amp design for a state variable filter is the better choice.

Project Simulation:

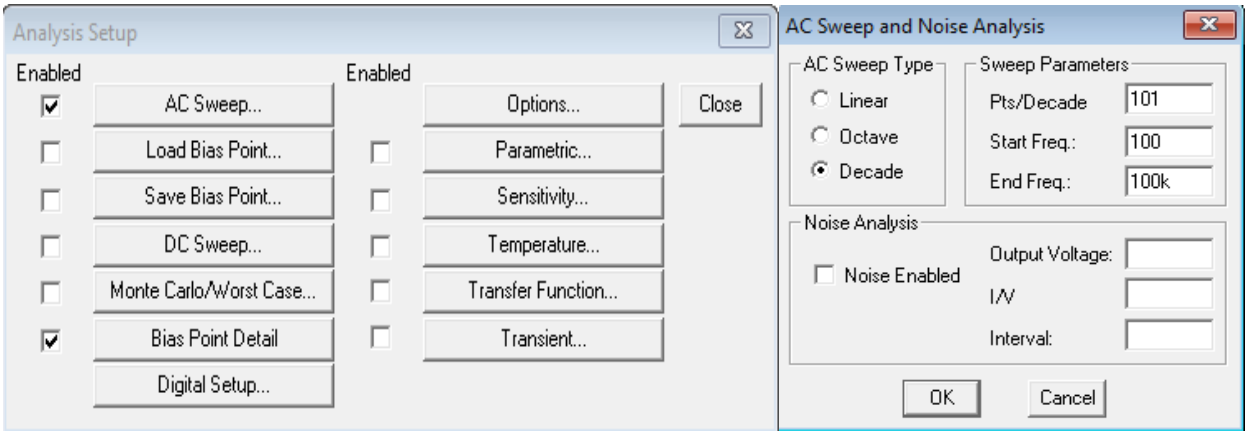
(1) 2 Op-Amp Configuration:

Schematics:

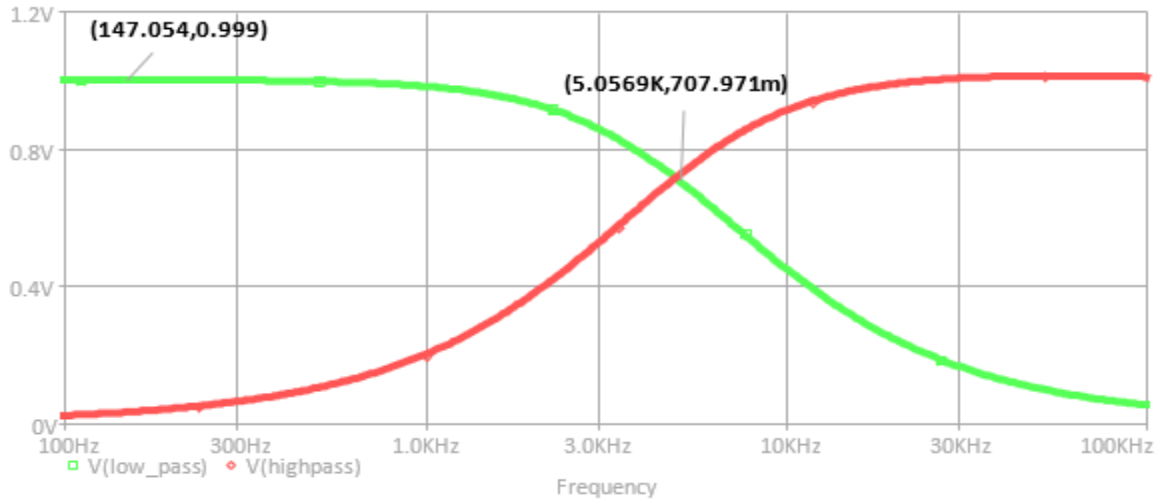


Here, R_1 and R3 are to be varied to change the gain. Since integrator has R = 10k ohm and C = 3.18 nF, cutoff frequency should be $\frac{1}{2\pi\times10\times10^{3}\times3.18\times10^{-9}}$ = 5 kHz

Analysis Setup:

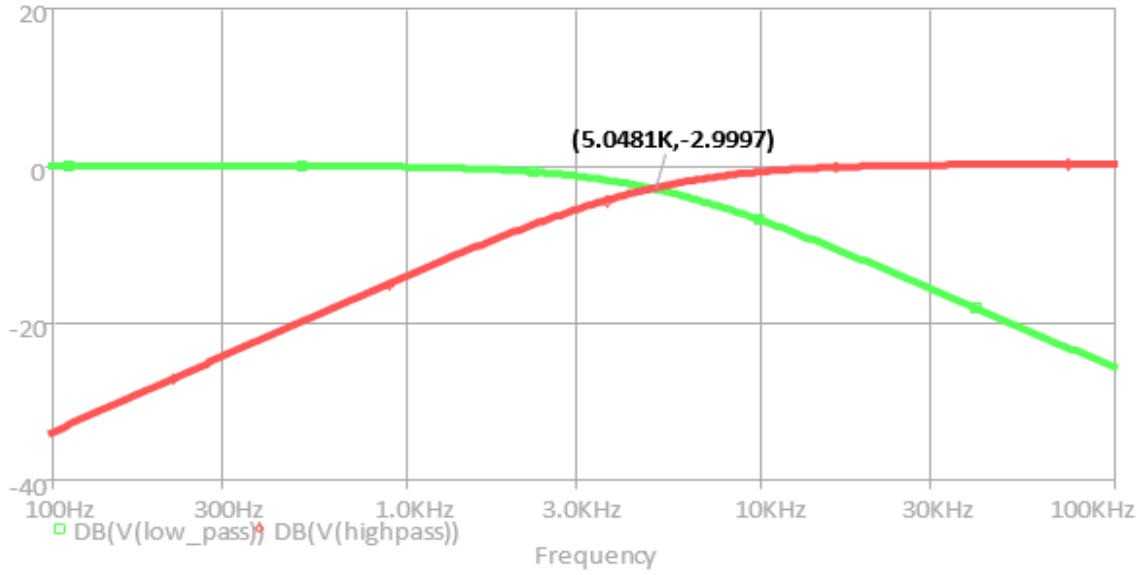


Gain vs Frequency:



Cutoff frequency is almost 5 kHz ; just as predicted. Gain is 1.

dB Gain vs Frequency:

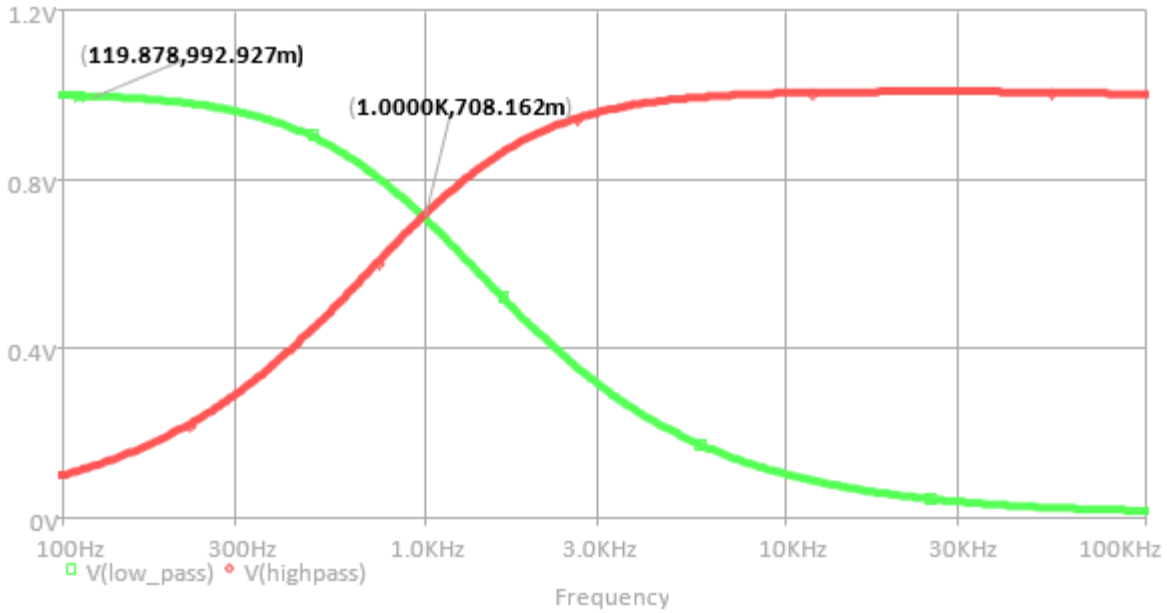


Gain is -3 dB at cutoff.

Frequency Control:

We can change the frequency by changing the resistor value in the integrator. Gain shouldl be unchanged since R_1 and R_3 are not changed.

If R is set to 50k ohm, then $f_C = \frac{1}{2\pi \times 50 \times 10^3 \times 3.18 \times 10^{-9}} = 1$ kHz

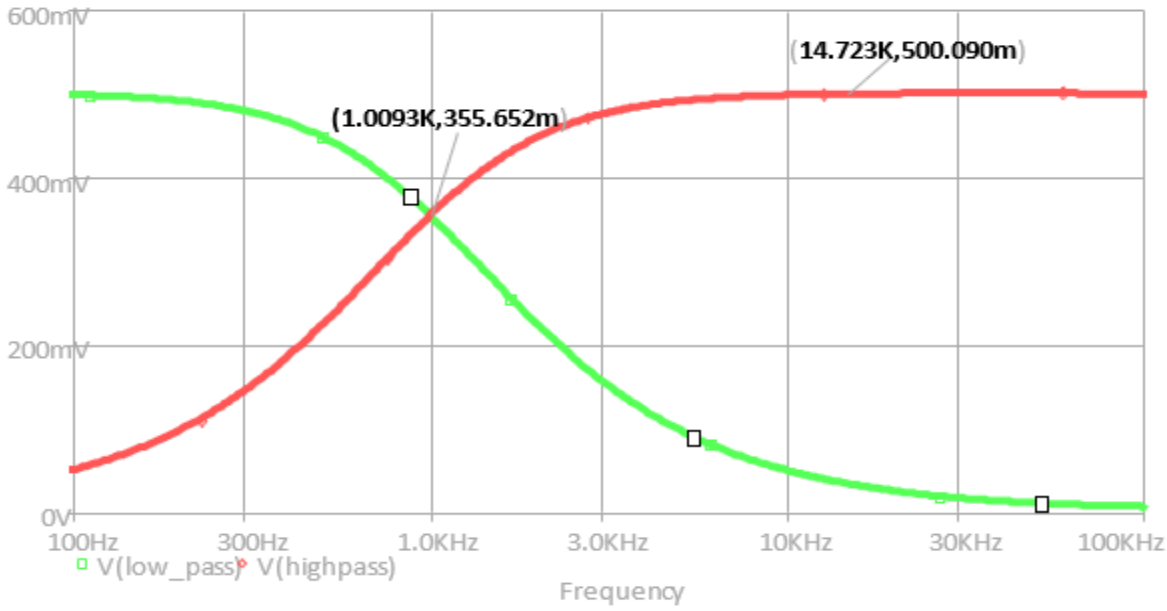


Cutoff frequency is 1 kHz and gain is almost unchanged.

Gain Control:

Now we will keep the cutoff at 1 kHz and change the gain by changing R_1 and R_3 and see if it has affected cutoff frequency or not.

R_1 and R_3 are changed to 2k ohm from 1k ohm.



Gain becomes half (0.5) of before, but cutoff frequency is fixed at 1 kHz.

Observation:

(1) Gain and cutoff frequency are not dependent on each other.

(2) Increasing gain controlling resistance (R_1 and R_3) will reduce the gain and vice versa.

(3) Only high pass and low pass response are present.

(4) It is a first order filter hence the graphs have 20 dB/decade slope at roll-off.

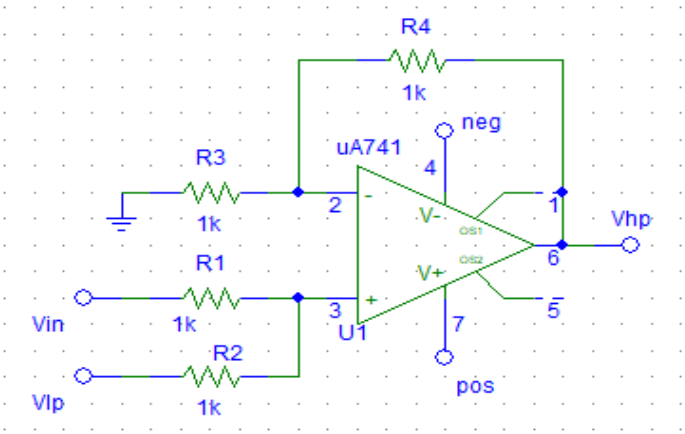
Theoretical Analysis:

(1) How is the low pass response created?

Ans: We found the low pass response by integrating the high pass response.

(2) How is the high pass response created?

Ans:



$$(V\text{-}) = V_{HP} \times \frac{R_3}{R_3+R_4} \quad \ldots\ldots(1)$$

$$\frac{V_{IN}-(V+)}{R_1} = \frac{(V+)-V_{LP}}{R_2}$$

$$\Rightarrow (V+) = \frac{V_{IN}*R_2+V_{LP}*R_1}{R_1+R_2} \quad \ldots\ldots\ldots(2)$$

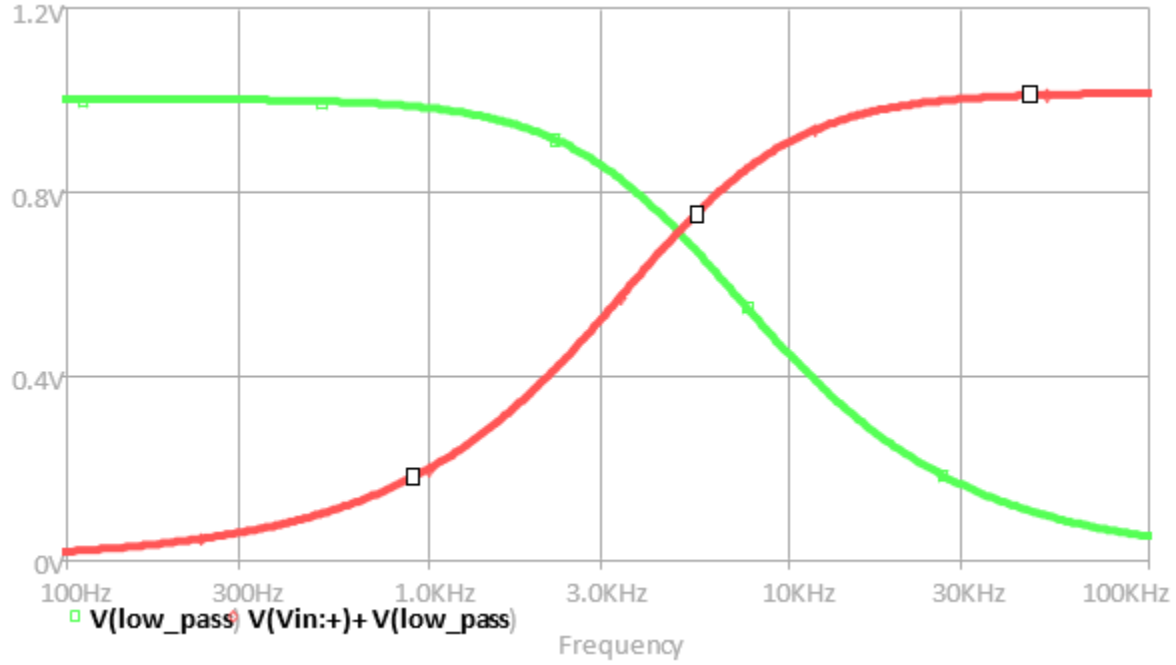
From (1) and (2),

$(V+) = (V\text{-})$

$$\Rightarrow V_{HP} = V_{IN} \times \frac{R_2(R_3+R_4)}{R_3(R_1+R_2)} + V_{LP} \times \frac{R_1(R_3+R_4)}{R_3(R_1+R_2)}$$

If $R_1 = R_2 = R_3 = R_4 = 1k$, then $\mathbf{V_{HP} = V_{IN} + V_{LP}}$

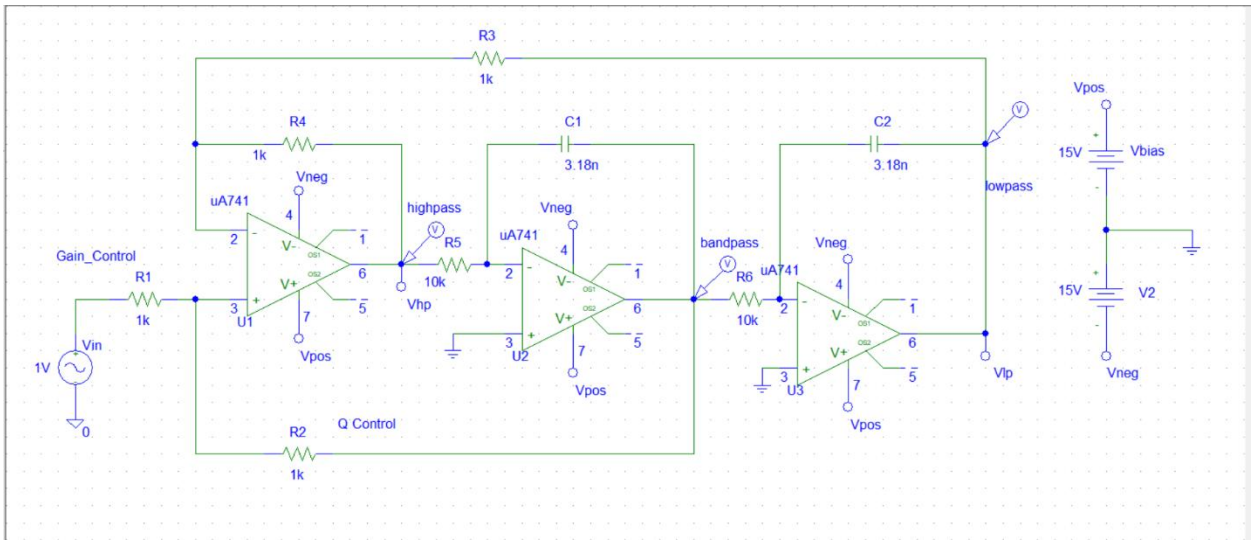
Let's see if our calculation is correct.



As can be seen from the graph that by adding input signal and low pass signal results in such a signal which is identical to high pass signal. So, our calculation is correct.

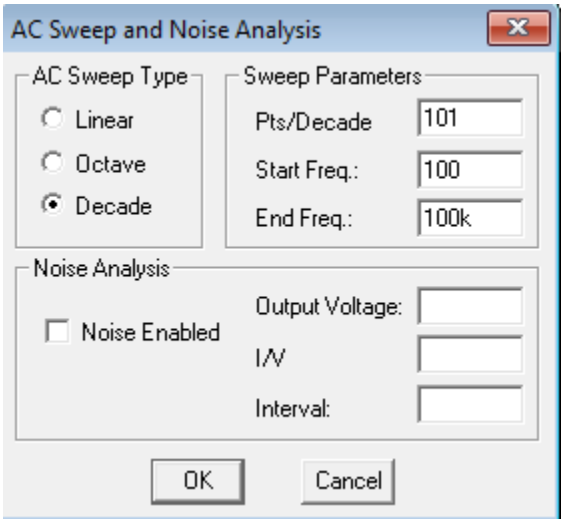
(2) 3 Op-Amp Configuration:

Schematics: (Input at non-inverting terminal)

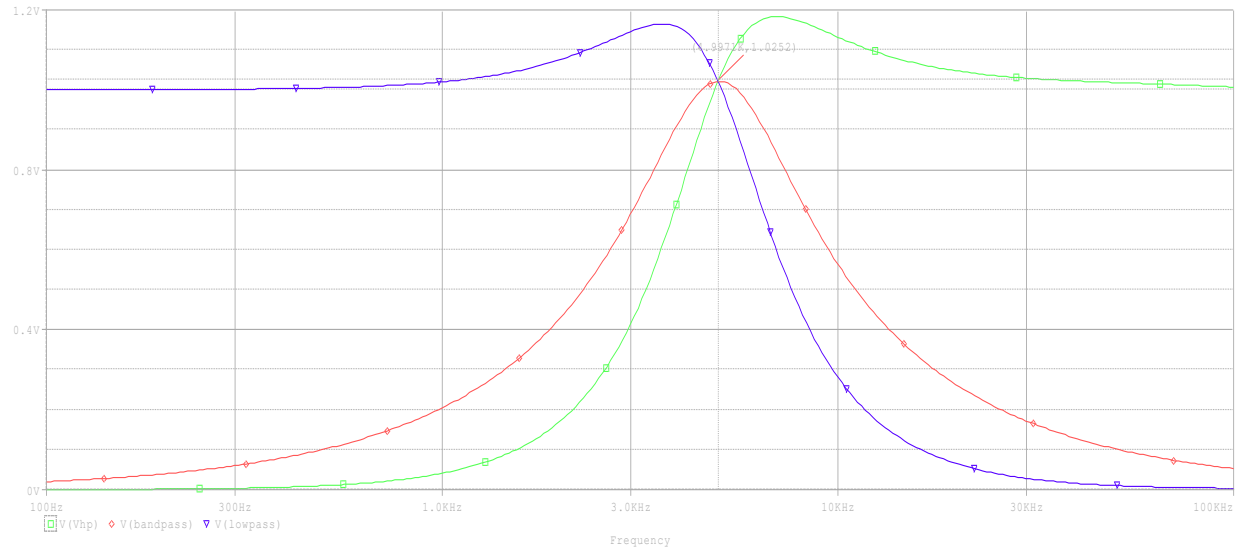


Here, R_1 and R2 are to be varied to change the gain and Q factor respectively. Since the integrators have R = 10k ohm and C = 3.18 nF, corner frequency should be $\frac{1}{2\pi\times10\times10^{3}\times3.18\times10^{-9}}$ = 5 kHz

Analysis Setup:

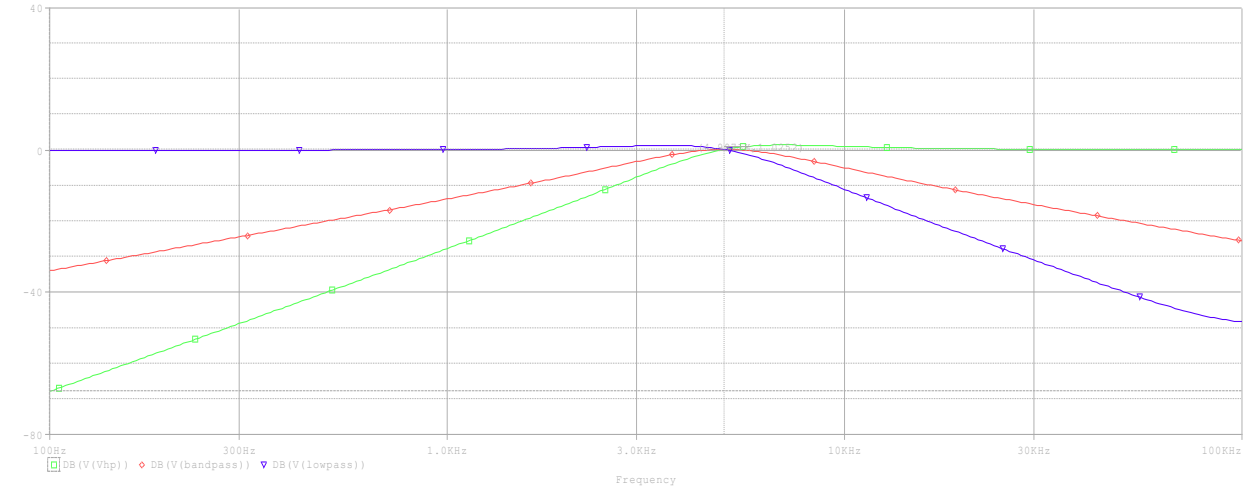


Gain vs Frequency:



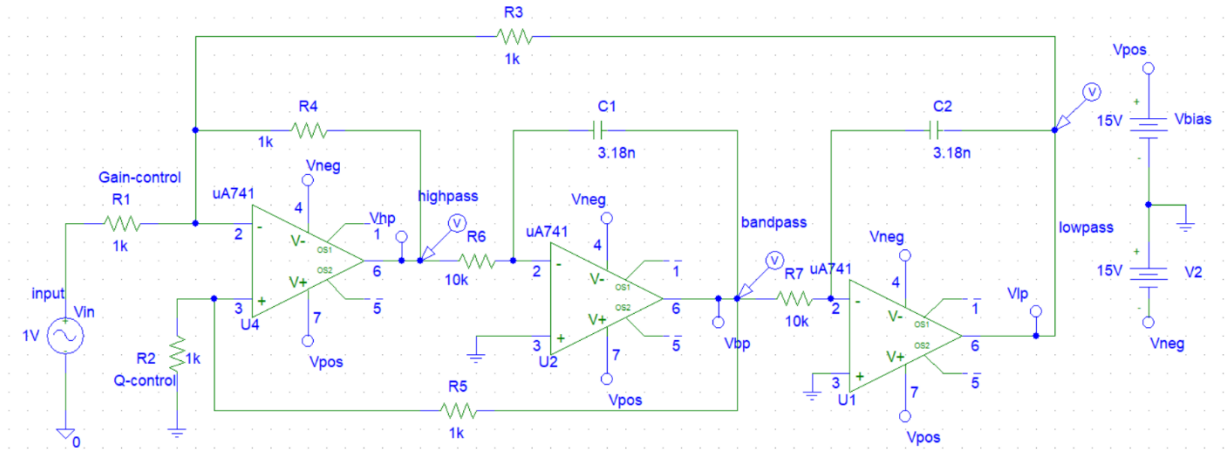
Passband gain of highpass/lowpass filter and peak gain of band pass filter is almost same (close to 1) which means Q is 1. Bandwidth is 5 kHz. Corner frequency is 5 kHz.

dB Gain vs Frequency:



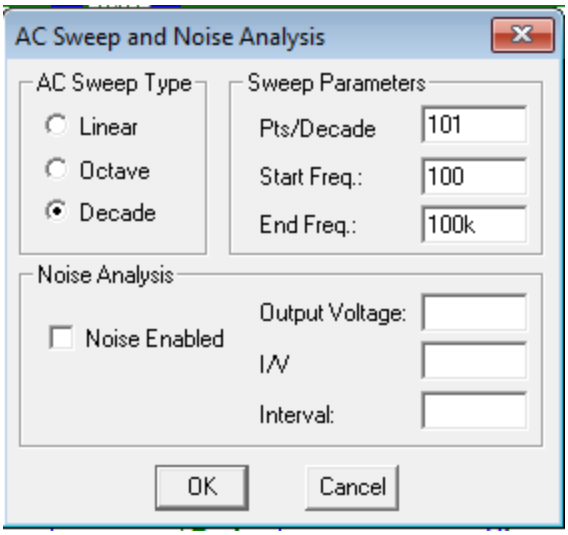
High pass and low pass filter has 40 dB/decade slope while band ass signal have 20 dB/decade slope.

Schematics: (Input at inverting terminal)

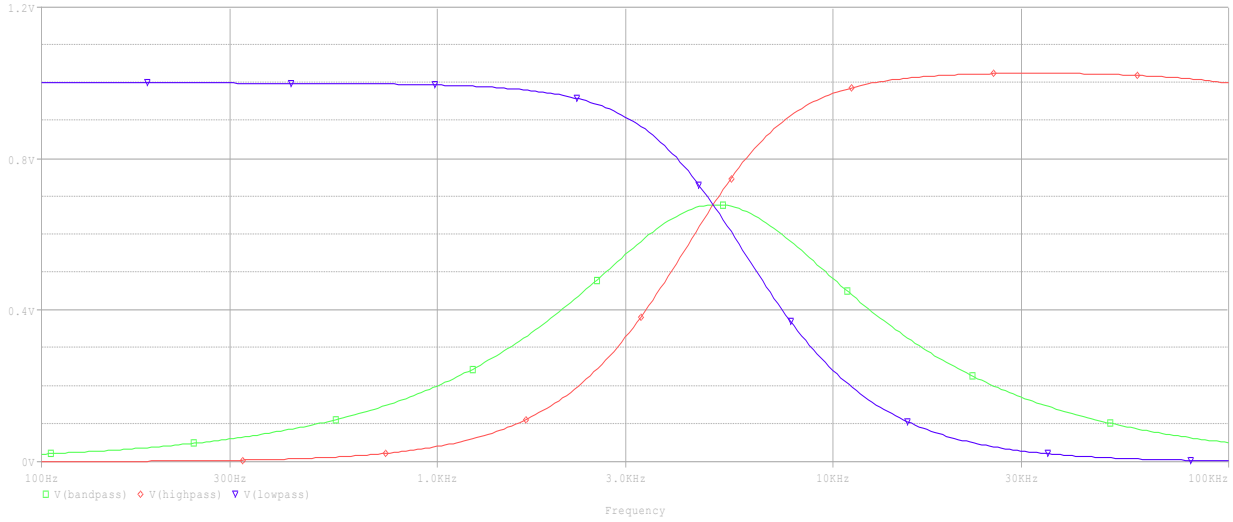


Here, R_1 and R2 are to be varied to change the gain and Q factor respectively. Since the integrators have R = 10k ohm and C = 3.18 nF, corner frequency should be $\frac{1}{2\pi\times10\times10^{3}\times3.18\times10^{-9}}$ = 5 kHz

Analysis Setup:

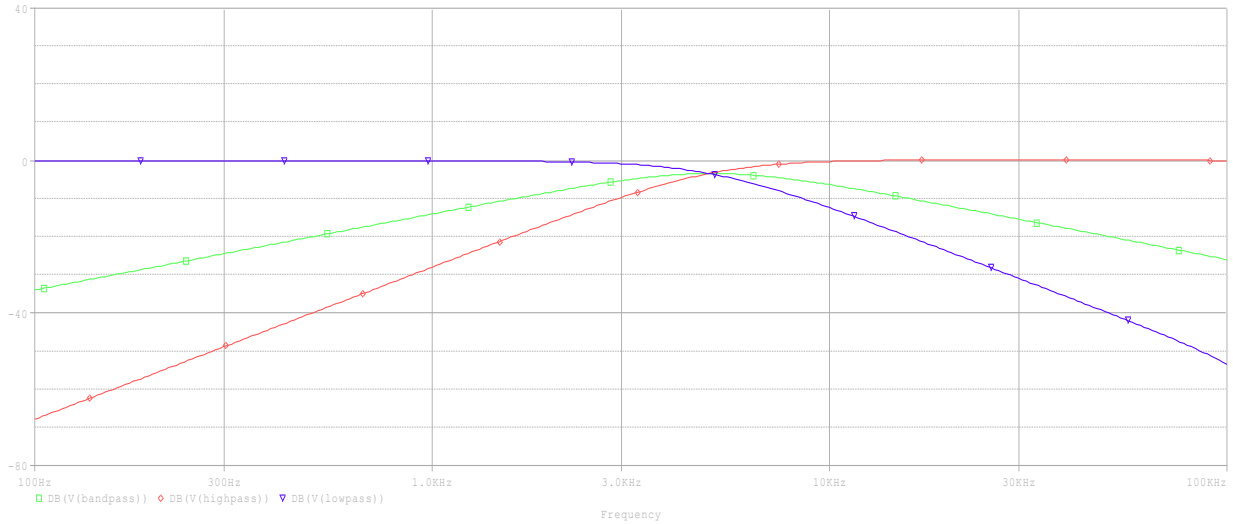


Gain vs Frequency:



Passband gain is almost 1 while the peak gain of and pass signal is 0.678. Corner frequency is 5 kHz.

dB Gain vs Frequency:

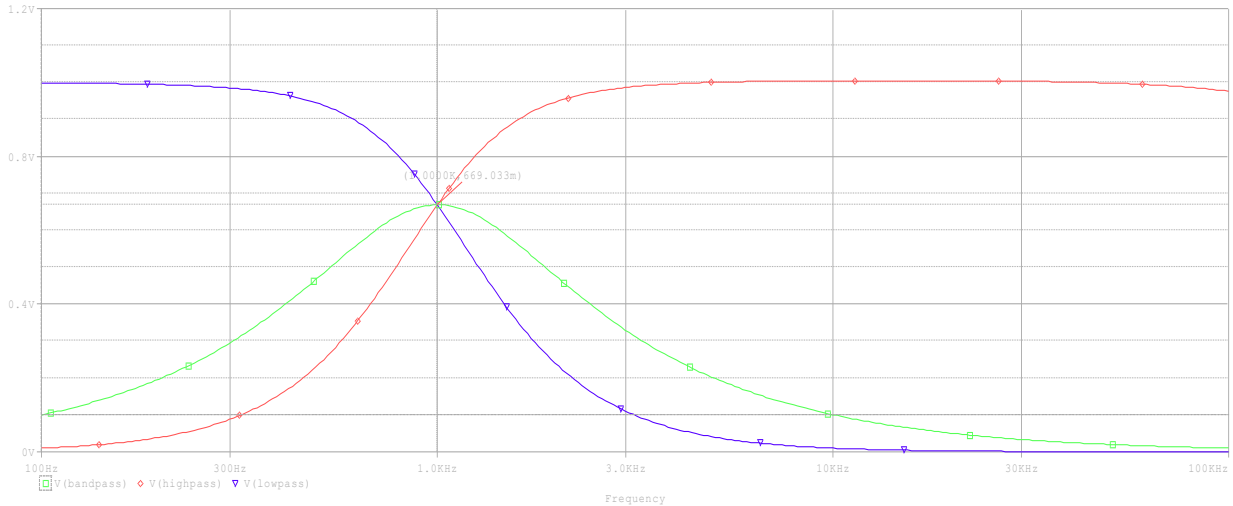


Bandwidth is (10.09-2.53)kHz or 7.56 kHz

Frequency Control:

We can change the frequency by changing the resistor value in the integrators. Gain and Q factor should be unchanged since R_1 and R_2 are not changed.

If R6 and R7 are set to 50k ohm, then $f_C = \frac{1}{2\pi \times 50 \times 10^3 \times 3.18 \times 10^{-9}} = 1$ kHz

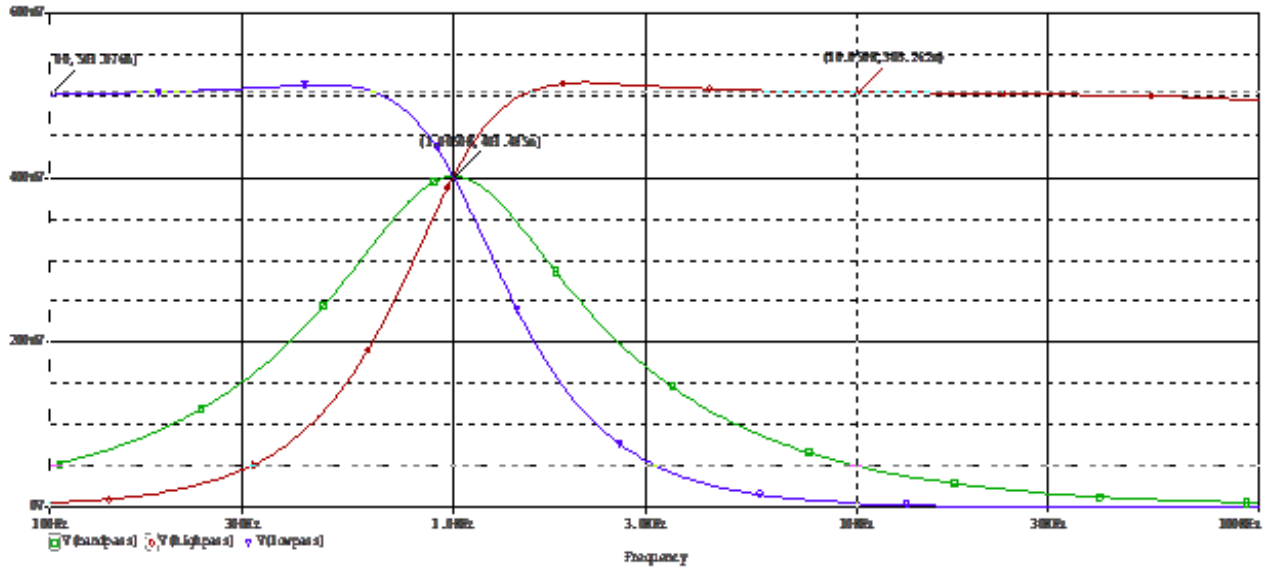


Only corner frequency changes from 5 kHz to 1 kHz. Passband gain, peak gain and Q are unchanged. Bandwidth is 1.5 kHz.

Gain Control:

Now we will keep the corner frequency at 1 kHz and change the gain by changing R_1 and see if it has affected other parameters.

R_1 is changed to 2k ohm from 1k ohm.



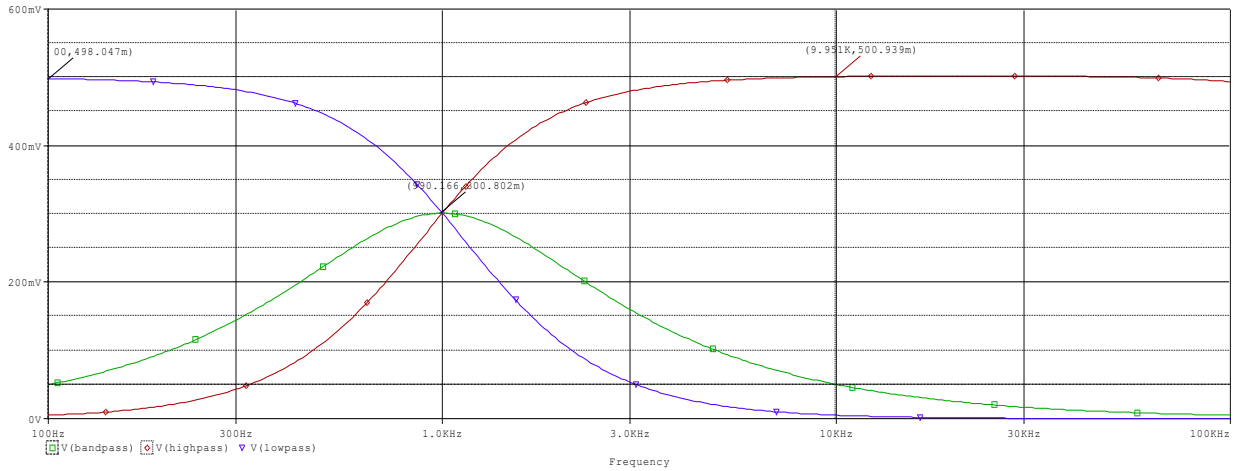
Corner frequency is unchanged.

Passband gain becomes 0.5 (half of previous value).
Bandwidth becomes 1.26 kHz. That means Q also changes which indicates this design is still not perfect.

Q-Factor Control:

Now we will change the Q-factor by changing R_2 and see if it has affected other parameters.

R_2 is changed to 2k ohm from 1k ohm.



Frequency and passband gain are not affected. Only the peak gain reduces to 0.3 from 0.401, so Q-factor has been changed.

Observations:

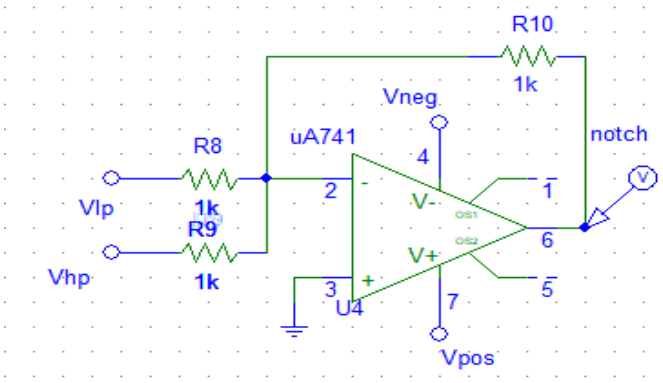
(1) Corner frequency is completely independent of others.

(2) Changing Q does not affect other parameters.

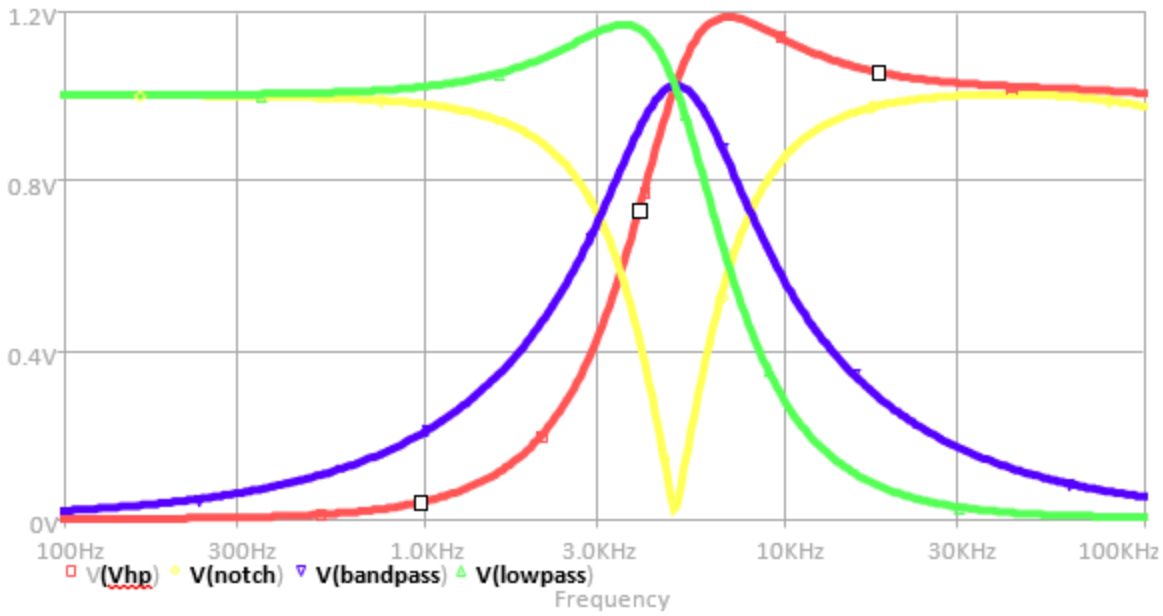
(3) Changing passband gain slightly changes Q. (This issue does not occur in 4 op-amp design).

Obtaining Notch Response:

We need to collect the high pass and low pass response through an inverting summer.



Simulation:



Theoretical Analysis:

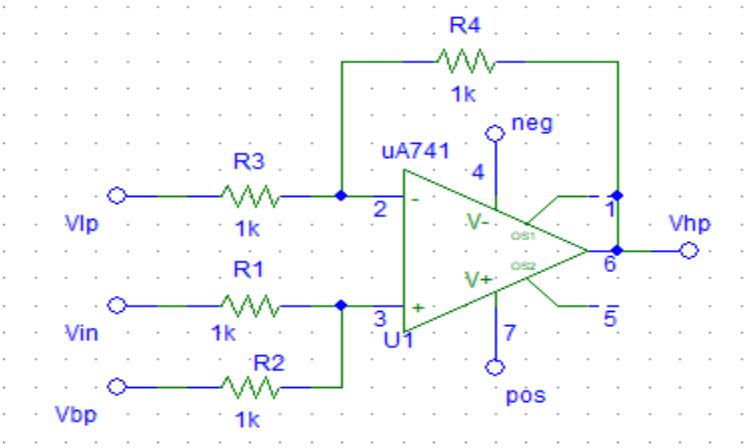
(1) How do we get band pass and low pass signal?

Ans: We get band pass and low ass signal by integrating high pass signal once and twice respectively.

(2) How is high pass signal produced?

Ans: Band pass and low pass feedback along with the input signal make the high pass signal.

(For input at non-inverting terminal)



$$\frac{V_{IN}-(V+)}{R_1} = \frac{(V+)-V_{LP}}{R_2}$$

$$\Rightarrow (V+) = \frac{V_{IN}*R_2+V_{LP}*R_1}{R_1+R_2} \quad \ldots\ldots(1)$$

$$\frac{V_{HP}-(V-)}{R_4} = \frac{(V-)-V_{LP}}{R_3}$$

$$\Rightarrow (V-) = \frac{V_{HP}*R_3+V_{LP}*R_4}{R_3+R_4} \quad \ldots\ldots(2)$$

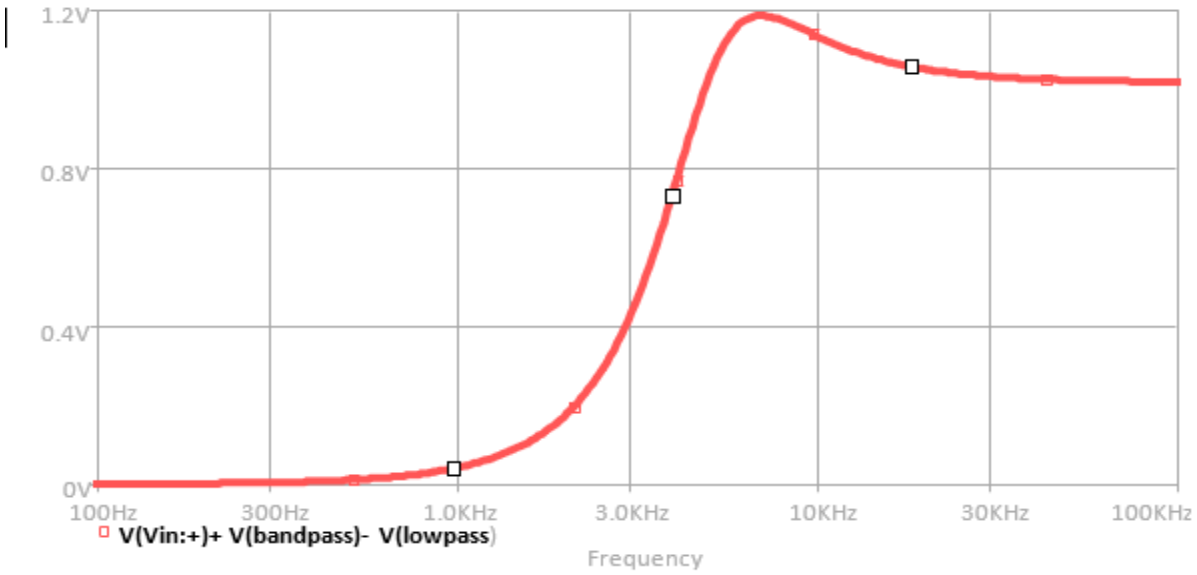
From (1) and (2),

V(+) = V(-)

$$\Rightarrow V_{HP} = V_{IN} \times \frac{R_2(R_3+R_4)}{R_3(R_1+R_2)} + V_{BP} \times \frac{R_1(R_3+R_4)}{R_3(R_1+R_2)} - V_{LP} \times \frac{R_4}{R_3}$$

If $R_1 = R_2 = R_3 = R_4 = 1k$, then $\mathbf{V_{HP} = V_{IN} + V_{BP} - V_{LP}}$

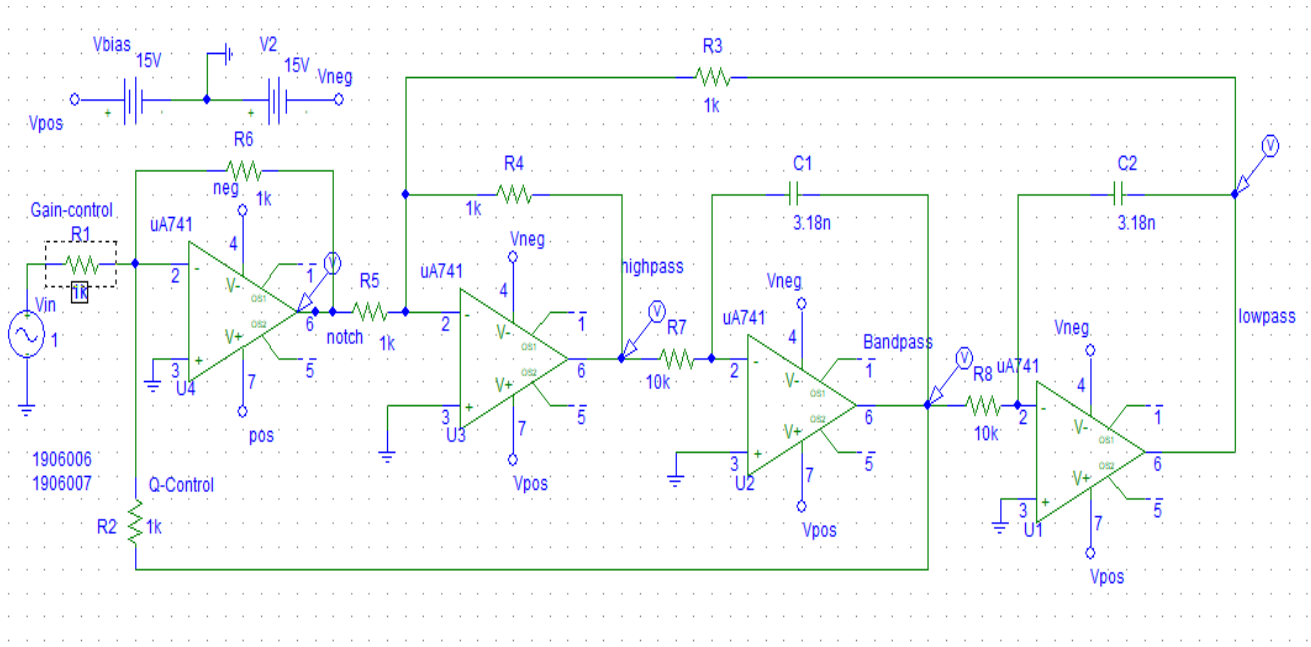
Let's see if our calculation is correct.



As can be seen from the graph that by adding input signal and band pass signal and then subtracting low pass signal results in such a signal which is identical to high pass signal. So, our calculation is correct.

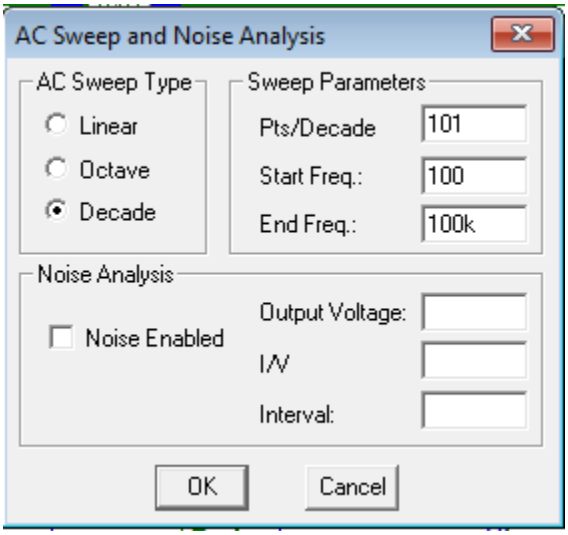
(3) 4 Op-Amp Configuration:

Schematics:

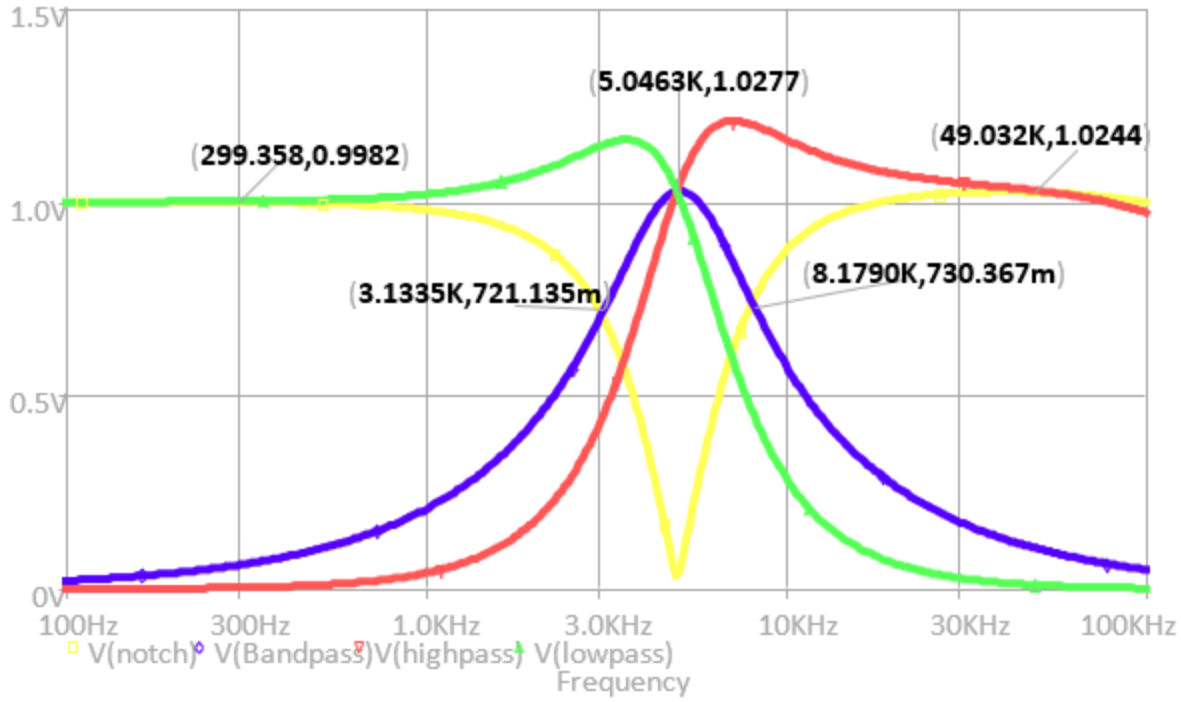


Here, R_1 and R2 are to be varied to change the gain and Q factor respectively. Since the integrators have R = 10k ohm and C = 3.18 nF, corner frequency should be $\frac{1}{2\pi\times10\times10^{3}\times3.18\times10^{-9}}$ = 5 kHz

Analysis Setup:

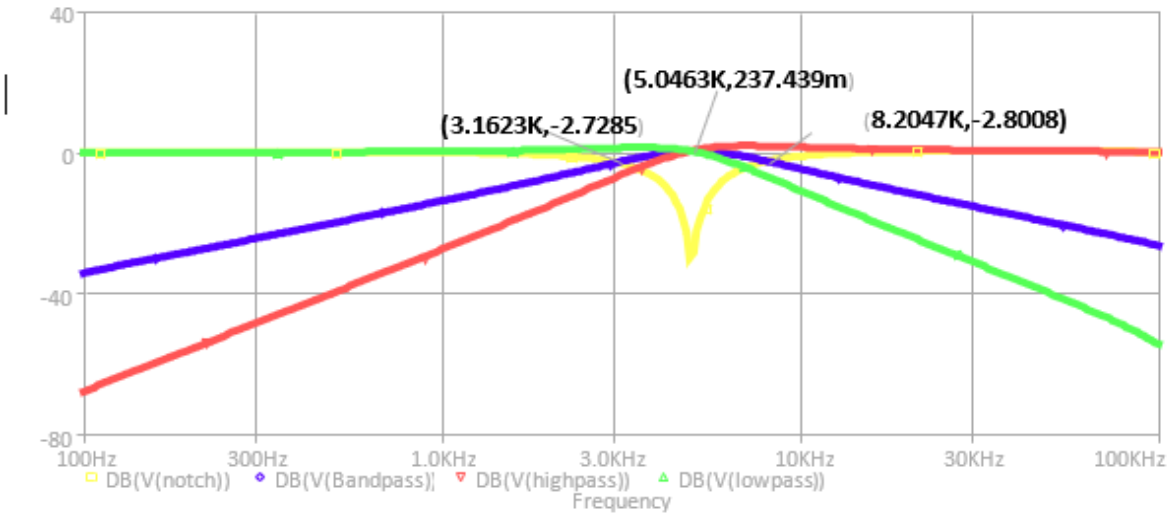


Gain vs Frequency:



Corner frequency is 5 kHz as expected. Passband gain is 1 and peak gain is 1.0277.

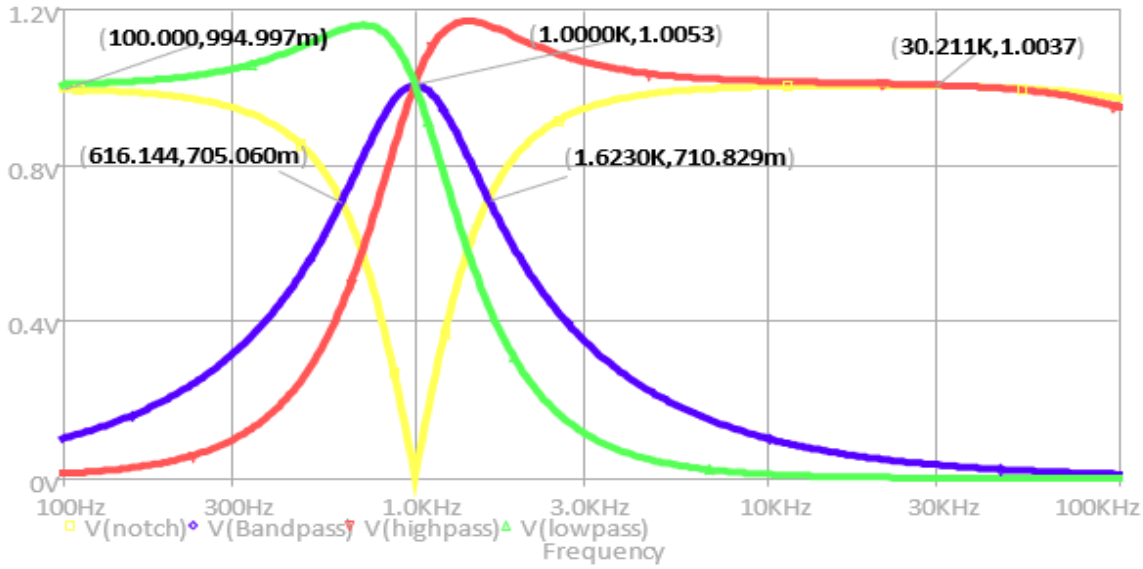
dB Gain vs Frequency:



Bandwidth is almost 5 kHz.

Frequency Control: We can change the frequency by changing the resistor value in the integrators. Gain and Q factor should be unchanged since R_1 and R_2 are not changed.

If R_1 and R_2 are set to 50k ohm, then $f_C = \frac{1}{2\pi \times 50 \times 10^3 \times 3.18 \times 10^{-9}} = 1$ kHz

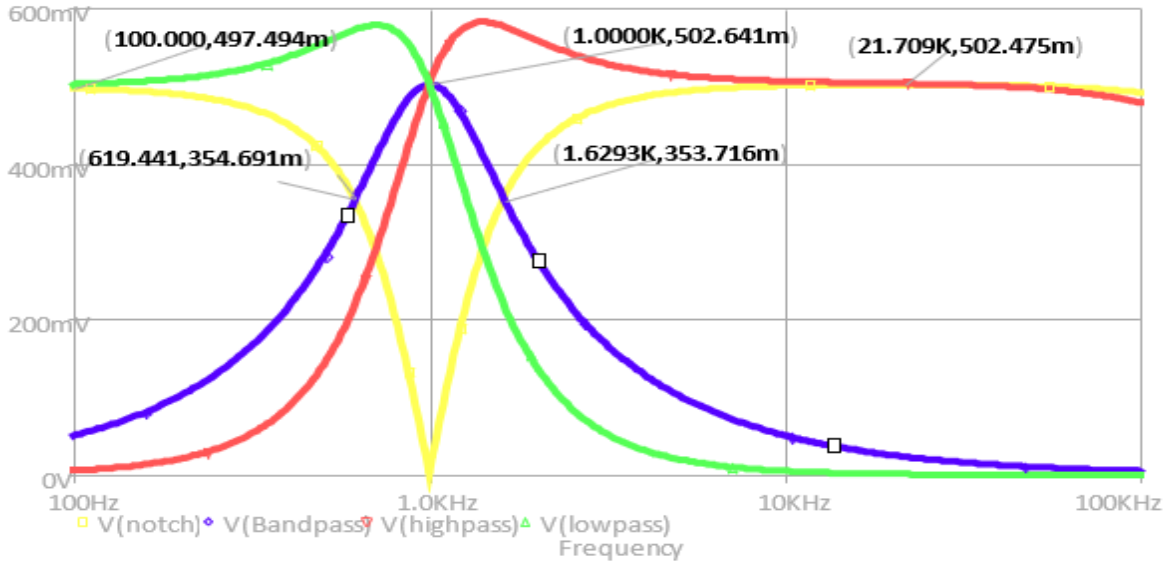


Only frequency changes from 5 kHz to 1 kHz. Passband gain and peak gain are unchanged. So, Q factor was not affected. Bandwidth is 1 kHz. So, Q is 1.

Gain Control:

Now we will keep the corner frequency at 1 kHz and change the gain by changing R_1 and see if it has affected other parameters.

R_1 is changed to 2k ohm from 1k ohm.



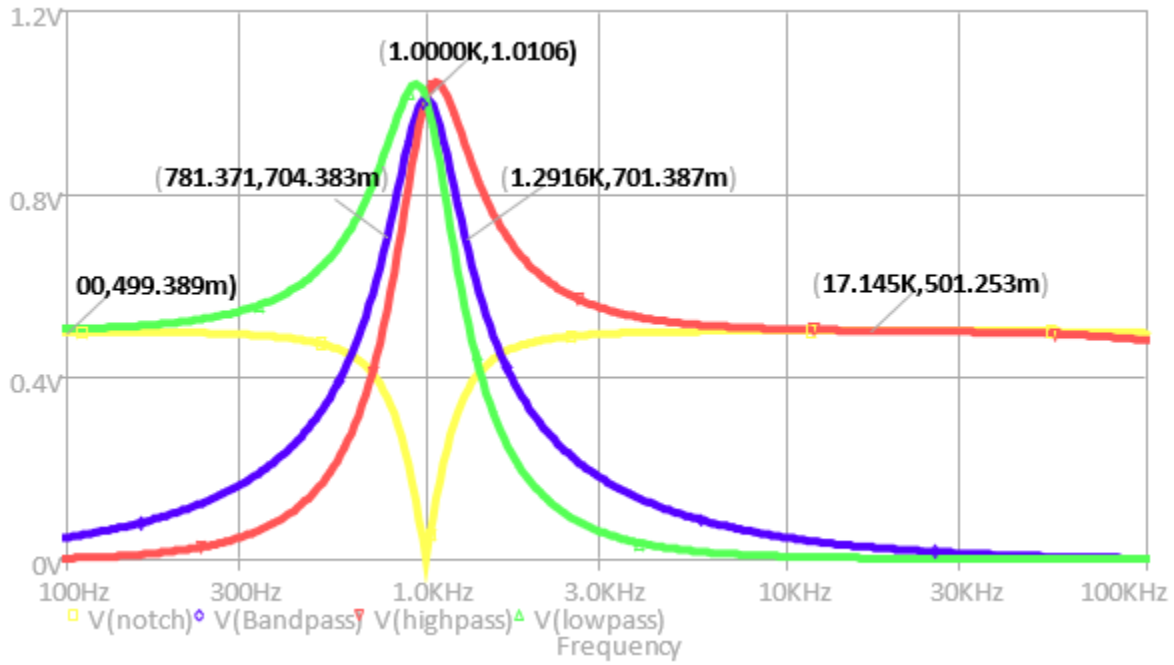
Corner frequency is unchanged. Passband gain becomes 0.5 (half of its previous value).

Bandwidth remains 1 kHz. So, Q factor is still 1. That means changing gain doesn't change Q.

Q-Factor Control:

Now we will change the Q-factor by changing R_2 and see if it has affected other parameters.

R_2 is changed to 2k ohm from 1k ohm.



Corner frequency and passband gain are unchanged. Peak gain becomes 1.0106 (double of previous value). That means Q is changed (increased by 100%)

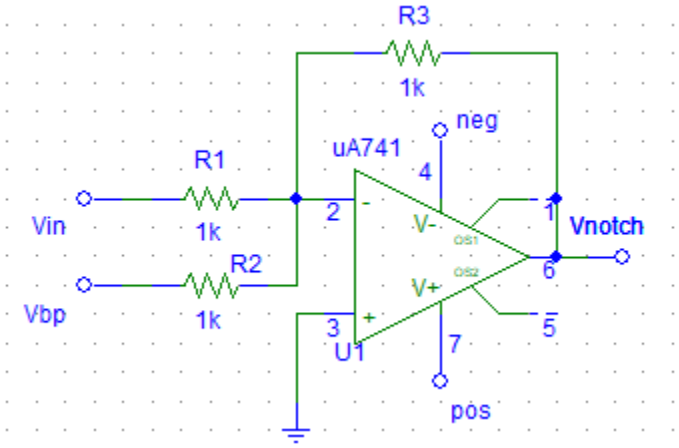
Observations:

Passband gain, Q factor and corner frequency are completely independent of each other in 4 op-amp configuration. Besides, all types of filter responses can be realized from a single circuit. So, state variable filter designing with 4 op-amp is the better choice.

Theoretical Analysis:

(1) How do we get the notch signal?

Ans: Input signal along with the band pass feedback make up the notch signal.

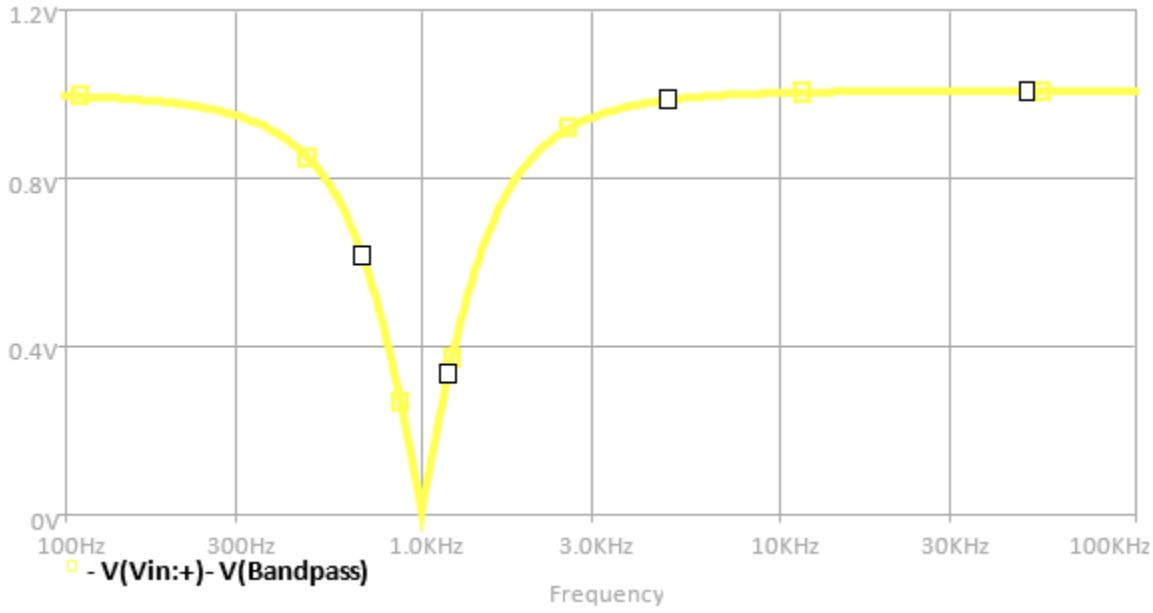


$$\frac{V_{IN}}{R_1} + \frac{V_{BP}}{R_2} + \frac{V_{notch}}{R_3} = 0$$

$$\Rightarrow V_{notch} = -V_{IN} \times \frac{R_3}{R_1} - V_{BP} \times \frac{R_3}{R_2}$$

If $R_1 = R_2 = R_3 = 1k$, then $V_{notch} = \mathbf{- V_{IN} - V_{BP}}$

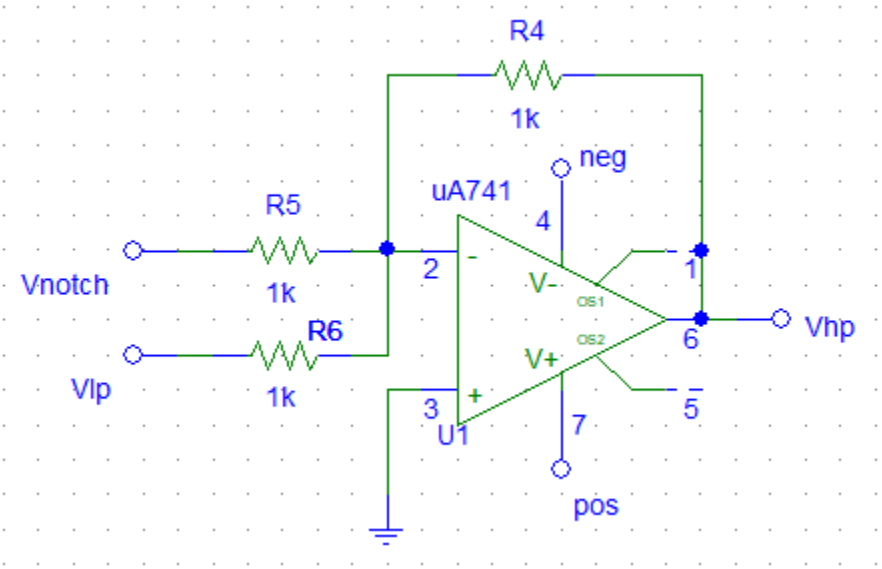
Let's see if our calculation is correct.



As can be seen from the graph that by adding input signal and band pass signal and then inverting the sum results in such a signal which is identical to notch signal. So, our calculation is correct.

(2) How is high pass signal produced?

Ans: Notch signal and low pass signal are added by an inverting summer and that produces the high pass signal.

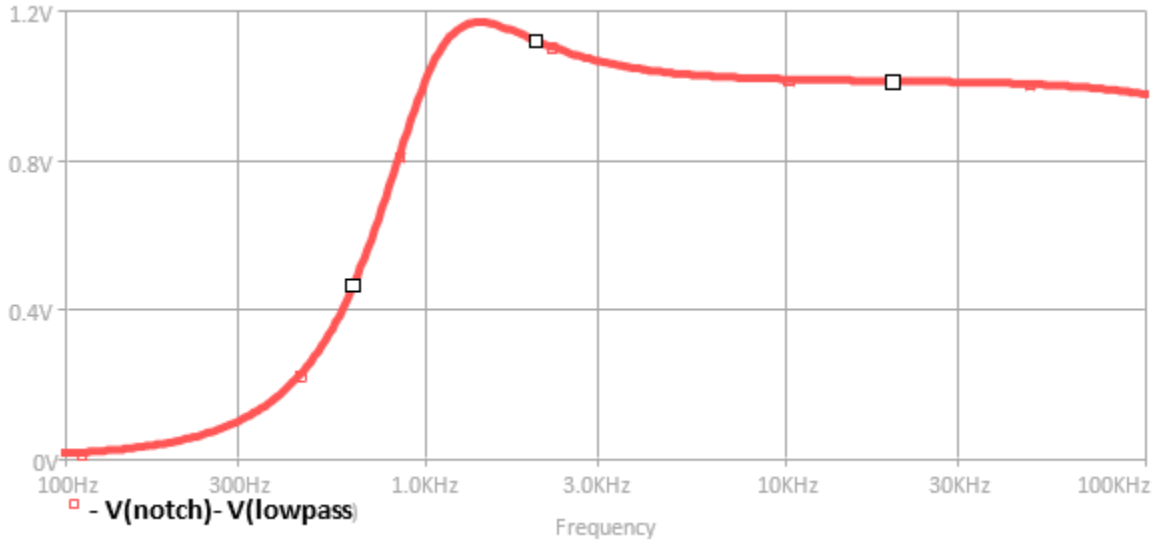


Here, $R_4 = R_5 = R_6 = 1k$ [by default and fixed]

So, $V_{notch} + V_{LP} + V_{HP} = 0$

$\Rightarrow V_{HP} = - V_{notch} - V_{LP}$

Let's see if our calculation is correct.



As can be seen from the graph that by adding notch signal and low pass signal and then inverting the sum results in such a signal which is identical to high pass signal. So, our calculation is correct.

Limitations of the Project:

State variable filter does not give proper output for high Q and very high frequency. It works better for Q<10 and corner frequency<100 kHz.

Scope for Future Development:

We might develop this project further in future by bringing some change in the design so that this filter becomes operational in very high frequency region.

Conclusion:

State variable filter can produce all types of filter response from a single circuit. It does so with the help of multiple feedback. Thus, state variable filter is very popular among active filters. In this project, we have implemented the state variable circuits in PSpice and run necessary simulations to check if we are achieving the properties and parameters for which state variable filter is known for. Considering the above simulations and results, we can say that it is possible to control corner frequency, Q factor and gain independently in a state variable filter. Besides, we have got all types of filter responses from a circuit which is a huge advantage. Considering this, we can say we have been able to achieve the objective in this project.